

Parallel and distributed Computing

Lecture 15

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Heterogeneity in GPU Computing

Heterogeneity refers to the use of different types of processing units (e.g., CPUs, GPUs, FPGAs (Field-Programmable Gate Arrays)) in a system to optimize performance and efficiency.

Heterogeneous Systems

1.CPU-GPU Systems:

1. CPUs handle sequential tasks and control flow, while GPUs handle parallel tasks.
2. *Example:* A machine learning system where the CPU preprocesses data and the GPU trains the model.

2.Multi-GPU Systems:

1. Multiple GPUs work together to solve large-scale problems.
2. *Example:* Deep learning training on multiple GPUs.

3.CPU-GPU-FPGA Systems:

1. Combines CPUs, GPUs, and FPGAs for specialized tasks.
2. *Example:* Real-time data processing in autonomous vehicles.

Challenges

- Programming Complexity:** Writing code for heterogeneous systems requires expertise in multiple programming models.
- Data Transfer Overhead:** Moving data between different processing units can be costly.
- Load Balancing:** Distributing tasks efficiently across heterogeneous resources.

Interconnection topologies define how components (e.g., SMs, memory) in a GPU are connected. The topology affects the performance, scalability, and efficiency of the GPU.

Common Interconnection Topologies

1. Mesh Topology:

1. Components are connected in a grid-like structure.
2. *Advantages*: Scalable and flexible.
3. *Disadvantages*: Higher latency for distant components.
4. *Example*: NVIDIA Volta architecture.

1. Ring Topology:

1. Components are connected in a circular fashion.
2. *Advantages*: Simple and low latency for neighboring components.
3. *Disadvantages*: Limited scalability.
4. *Example*: AMD RDNA architecture.

2. Crossbar Topology:

1. Each component is connected to every other component via a switch.
2. *Advantages*: Low latency and high bandwidth.
3. *Disadvantages*: High cost and complexity.
4. *Example*: High-end GPUs for data centers.

1. Fat Tree Topology:

1. A hierarchical topology with multiple levels of switches.
2. *Advantages*: Scalable and efficient for large systems.
3. *Disadvantages*: Complex to design and manage.
4. *Example*: GPUs in supercomputers.

GPU Architectures

- **NVIDIA Tesla V100:** Designed for AI and high-performance computing.
- **AMD Radeon RX 6900 XT:** Optimized for gaming and general-purpose computing.

Heterogeneous Systems

- **NVIDIA DGX Systems:** Combine multiple GPUs and CPUs for AI workloads.
- **AMD Instinct MI Series:** Designed for machine learning and data analytics.

Interconnection Topologies

- **NVIDIA NVLink:** High-speed interconnect for multi-GPU systems.
- **AMD Infinity Fabric:** Scalable interconnect for CPUs and GPUs.

Challenges in GPU Computing

- 1.Programming Complexity:** Writing efficient GPU code requires expertise in parallel programming.
- 2.Memory Management:** Optimizing data transfer between CPU and GPU memory.
- 3.Scalability:** Ensuring performance scales with the number of cores and GPUs.
- 4.Energy Efficiency:** Reducing power consumption in high-performance GPUs.

Kindly read the book before and after the lecture, if some terms/ words are not covered in the lecture, if appear in exam it might not appear strange or out of course.